Extrema and the Mean Value Theorem

Section 1: Critical Points & First Derivative Test

1 What are the critical points of $f(x) = x^3 - 3x^2 - 9x + 5$?

a. $x = -1$ and $x = 3$

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

2 By the First Derivative Test, what type of extremum does $f(x) = x^3 - 3x^2 - 9x + 5$ have at $x = -1$?

a. Local maximum because f' changes from positive to negative

b. Local minimum because f' changes from negative to positive

c. Inflection point

d. No extremum

3 Can a critical point occur at a value where $f'(x)$ is undefined?

a. Yes, provided x is in the domain of f

b. No, f' must always equal zero

c. Only if $f''(x) = 0$

d. Only for rational functions

Section 2: Mean Value Theorem (MVT)

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4 What are the conditions required to apply the Mean Value Theorem on $[a, b]$?

a. Continuous on $[a, b]$ and differentiable on (a, b)

b. Differentiable on $[a, b]$ and $f(a) = f(b)$

c. Continuous on (a, b) only

d. $f'(c) = 0$ on $[a, b]$

5 Find the value of c guaranteed by the MVT for $f(x) = x^2$ on the interval $[0, 4]$.

a. 2

b. 1

c. $\sqrt{2}$

d. 2.5

6 Why does the MVT NOT apply to $f(x) = |x|$ on $[-2, 2]$?

a. f is not differentiable at $x = 0$

b. f is not continuous on $[-2, 2]$

c. $f(-2) \neq f(2)$

d. The interval is not symmetric

Section 3: Rolle's Theorem

7 What additional hypothesis does Rolle's Theorem require compared to the MVT?

a. $f(a) = f(b)$

b. $f'(a) = f'(b)$

c. $f''(x) = 0$

d. $f(a) = 0$

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8 Verify Rolle's Theorem for $f(x) = x^2 - 5x + 4$ on $[1, 4]$. What is the value of c?

a. 2.5

b. 2.0

c. 3.0

d. 1.5

9 If a differentiable function has 3 distinct real zeros, what is the minimum number of zeros its derivative $f'(x)$ must have?

a. 2

b. 1

c. 3

d. 0

Section 4: Absolute Extrema on Closed Intervals (Candidates Test)

10 Find the absolute maximum value of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on $[-2, 3]$.

a. 8

b. 1

c. -19

d. -8

11 What is the absolute minimum value in the problem above?

a. -19

b. -8

c. -3

d. 1

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12 Where can the absolute extrema of a continuous function on a closed interval $[a, b]$ occur?

a. At critical points inside (a, b) or at the endpoints a, b

b. Only at endpoints

c. Only where $f'(x) = 0$

d. Anywhere on the interval

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